

Titanic Exploratory Data Analysis (EDA) Report

1. Project Title

Titanic Dataset Exploratory Data Analysis

2. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand the data, find patterns, handle missing values, and analyze factors affecting passenger survival.

3. Dataset Description

The dataset contains information about Titanic passengers including:

- Passenger ID
- Survival status
- Passenger class
- Name
- Gender
- Age
- Number of siblings/spouses
- Number of parents/children
- Ticket information
- Fare
- Cabin
- Embarked location

4. Tools and Libraries Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

5. Data Loading

The Titanic dataset was imported into Python using Pandas. Initial inspection was performed to understand the structure of the dataset.

6. Data Cleaning

The following steps were performed:

- Checked missing values
- Identified null values in Age, Cabin, and Embarked columns
- Handled missing values
- Removed unnecessary columns where required
- Prepared the dataset for analysis

7. Exploratory Data Analysis

The following analysis was performed:

Dataset Overview

- Displayed first few rows of the dataset
- Checked dataset shape
- Reviewed column information
- Generated statistical summary

Missing Value Analysis

Missing values were identified and analyzed to understand data quality.

Survival Analysis

Analyzed passenger survival based on:

- Gender
- Passenger class
- Age

8. Visualizations Created

Graphs were created to understand relationships in the dataset:

- Survival count plot
- Gender-wise survival analysis
- Passenger class survival analysis

- Age distribution
- Correlation analysis

9. Key Findings

- Female passengers had higher survival rates compared to male passengers.
- Passengers from higher classes had better survival chances.
- Age distribution showed that passengers belonged to different age groups.
- Some columns contained missing data which required preprocessing.

10. Conclusion

The Titanic dataset EDA helped in understanding passenger characteristics and survival patterns. Data cleaning and visualization provided useful insights about factors affecting survival.

11. Files Included

- Titanic_EDA.ipynb
- Titanic_EDA_Report.pdf
- README.md
- Dataset file (optional)